Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology
B.Tech. | Sem-5 | Winter-2025

Course Code	:	2501CS501	**Date**	:	17-10-2025
Course Name	:	Building RESTful APIs with ASP.NET Core	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain HTTP status codes with examples.	**4**
	(B)	List any three HTTP verbs and their purpose.	**3**
		OR	
		Define: JSON Serialization & Deserialization. Write it with an example.	
	(C)	Prepare a .NET Core Web API for managing a Student entity with the following requirements: implement endpoints for Create and Read operations. Clearly explain the steps involved, including model creation, DbContext configuration, and controller implementation.	**7**
		OR	
		Prepare a .NET Core Web API for managing a Student entity with the following requirements: implement endpoints for Update and Delete operations. Clearly explain the steps involved, including model creation, DbContext configuration, and controller implementation.	
Q.2	**(A)**	Explain synchronous and asynchronous queries in EF Core.	**4**
	(B)	Write a short note on Migrations & Scaffolding in EF Core.	**3**
		OR	
		List types of LINQ queries.	
	(C)	Describe LINQ query operators with examples.	**7**
		OR	
		Discuss the use of DbSet and DbContext in building APIs with an example.	
Q.3	**(A)**	Prepare a note on Attribute Routing with an example.	**4**
	(B)	Write a short note on API versioning.	**3**
		OR	
		Write a short note on Route Constraints.	
	(C)	Discuss Fluent Validation implementation in ASP.NET Core APIs.	**7**

OR

Explain different API versioning techniques with examples.

Q.4 **(A)** Prepare a jQuery example using the .on() method to handle multiple events (click, mouseenter, and mouseleave) on a button element. **4**

(B) Explain jQuery selectors with any two examples. **3**

OR

Explain $(document).ready() in jQuery.

(C) Define AJAX. Write a note on Calling AJAX Methods with an example. **7**

OR

Write a note on jQuery Form Events with an example.

Q.5 **(A)** Compare: Middleware and Action Filters. **4**

(B) Demonstrate Global Error Handling in Web API. **3**

OR

Demonstrate Role-Based Access Control (RBAC).

(C) Prepare a detailed note on implementing JWT Authentication with an example. **7**

OR

Prepare a note on pipeline of middleware execution in ASP.NET Core with app.Use(), app.Map() & app.Run() with an example.

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